Exercise 10.4 - AI4I Data Analysis

Table of contents

Introduction 1

Data Preparation 2

Basic Plots 2
- Lineplot . . . 2
- Histograms . . . 3
- Boxplots and Violinplots . . . 5

Relationships Between Variables 7
- Scatterplot . . . 7
- Pairplot . . . 8
- Correlation Heatmap . . . 9

Advanced Plots 11
- Jointplot . . . 11
- Stripplot . . . 12
- Countplot . . . 14

Observations 15
- Are there any obvious outliers in the data? . . . 15
- Which two variables seem most related? . . . 15
- Are there any non-linear patterns visible? . . . 15
- How are the distributions shaped? . . . 15

Introduction

In this report, the AI4I 2020 Predictive Maintenance Dataset is analyzed using Pandas, Matplotlib and Seaborn.

The dataset contains 10,000 measurements from a simulated industrial machine.

Data Preparation

```
# Importiert die Bibliotheken zur Datenverarbeitung und Visualisierung.
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Lädt den AI4I-Datensatz.
url = "https://archive.ics.uci.edu/static/public/601/ai4i+2020+predictive+maintenance+dataset
df = pd.read_csv(url)

# Behält die drei Messgrößen und die UDI.
df = df[["UDI", "Torque [Nm]", "Rotational speed [rpm]", "Air temperature [K]"]]

# Zeigt die ersten fünf Datensätze an.
df.head()
```

	UDI	Torque [Nm]	Rotational speed [rpm]	Air temperature [K]
0	1	42.8	1551	298.1
1	2	46.3	1408	298.2
2	3	49.4	1498	298.1
3	4	39.5	1433	298.2
4	5	40.0	1408	298.2

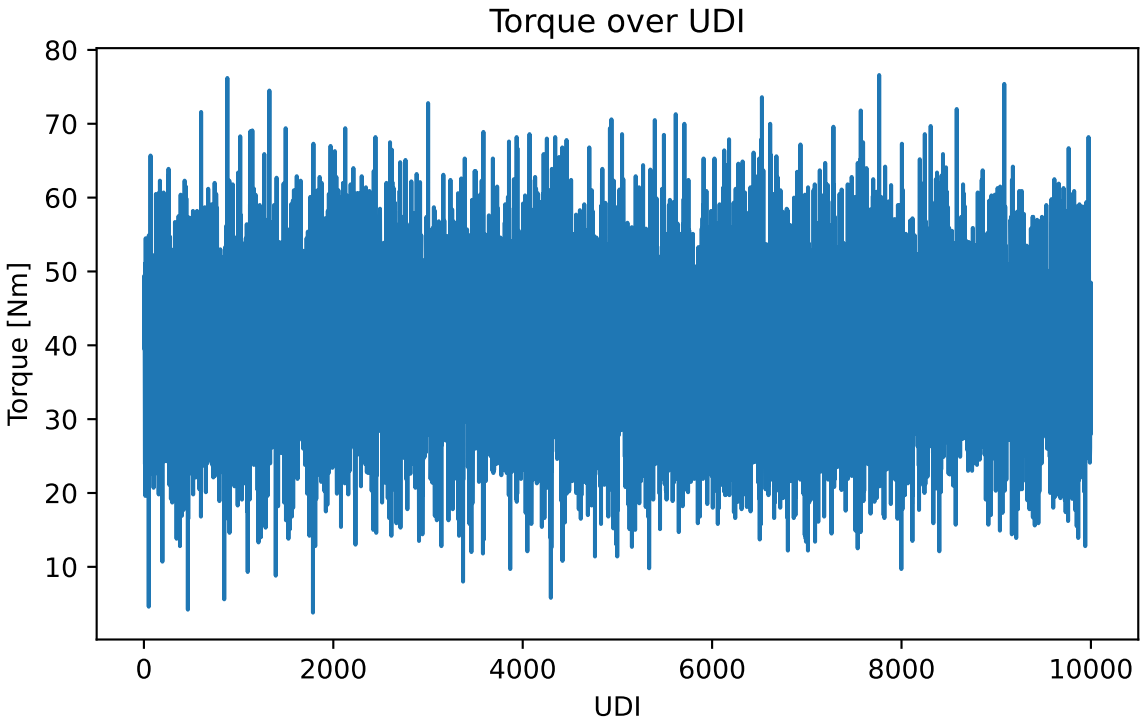
Basic Plots

Lineplot

```
# Erstellt ein Liniendiagramm des Drehmoments über der UDI.
plt.figure(figsize=(7, 4))
sns.lineplot(data=df, x="UDI", y="Torque [Nm]")

plt.title("Torque over UDI")
plt.xlabel("UDI")
```

```
plt.ylabel("Torque [Nm]")
plt.show()
```

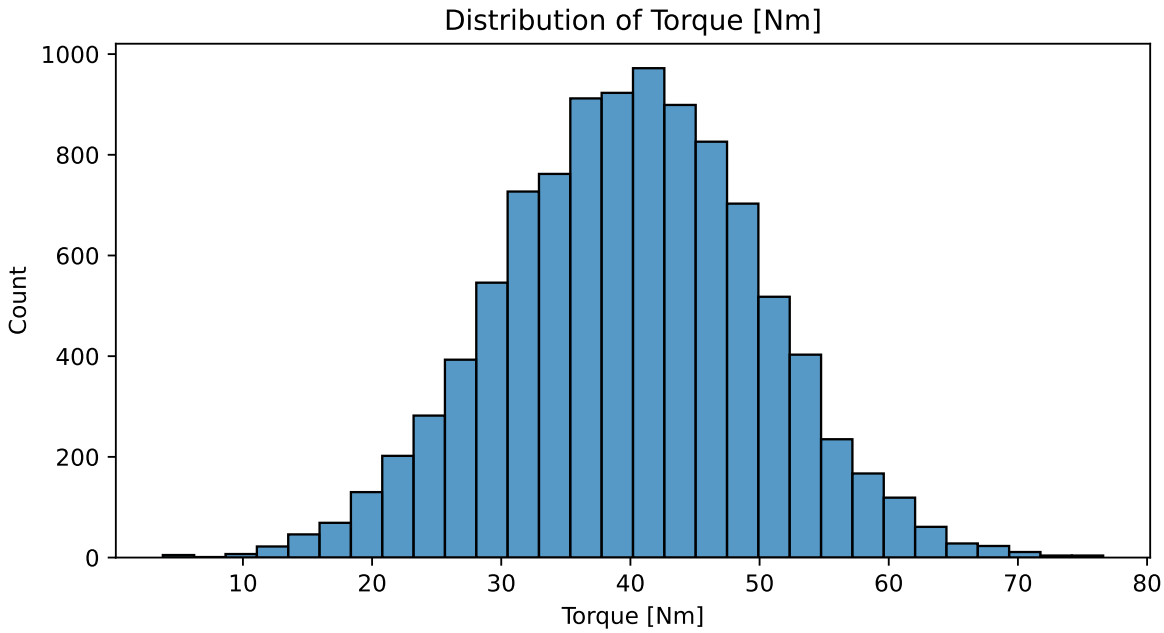


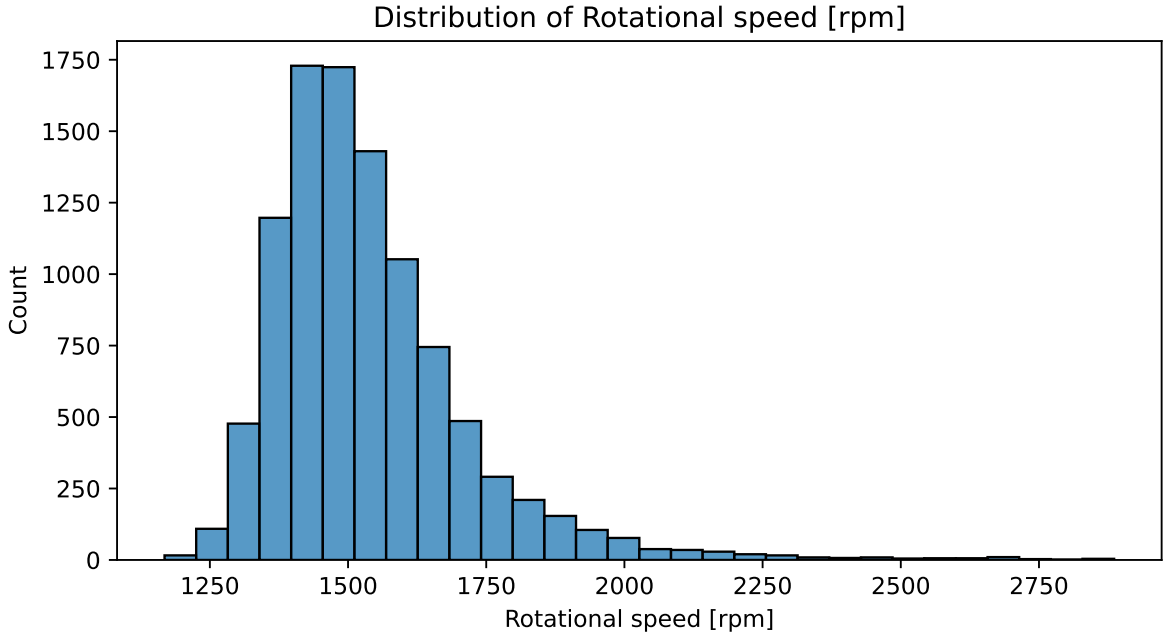
Histograms

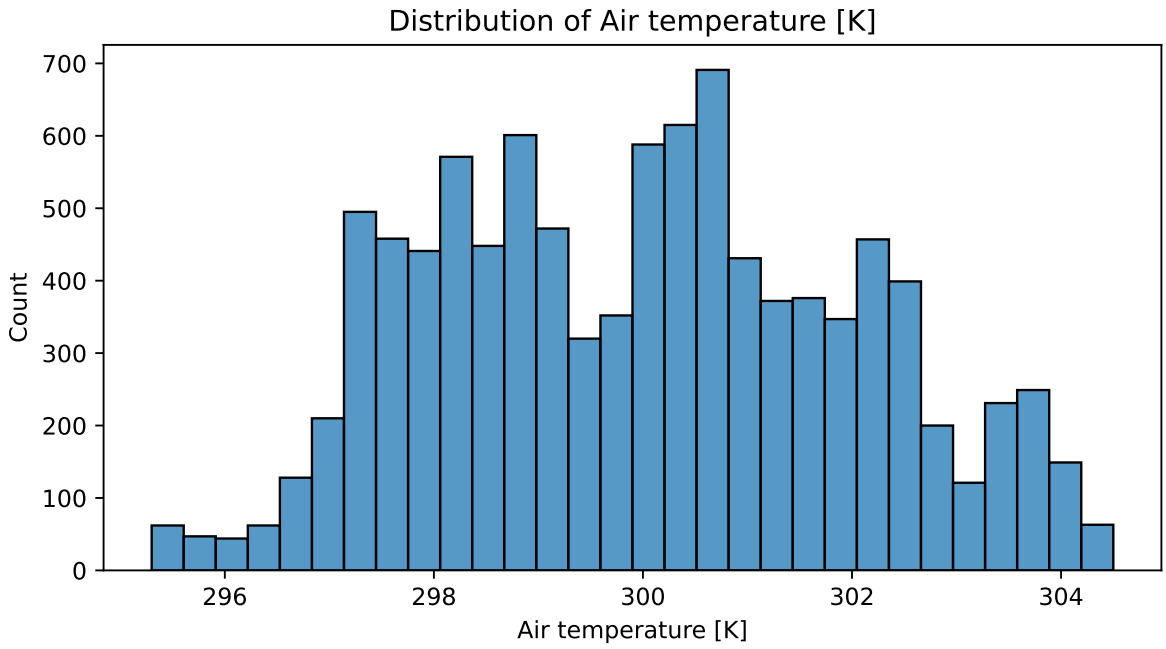
```
# Legt die drei Messgrößen fest.
columns = ["Torque [Nm]", "Rotational speed [rpm]", "Air temperature [K]"]

# Erstellt für jede Messgröße ein Histogramm.
for column in columns:
    plt.figure(figsize=(8, 4))
    sns.histplot(data=df, x=column, bins=30)

    plt.title(f"Distribution of {column}")
    plt.xlabel(column)
    plt.ylabel("Count")
    plt.show()
```


Distribution of Torque [Nm]
Count
Torque [Nm]
0
200
400
600
800
1000
10
20
30
40
50
60
70
80


Distribution of Rotational speed [rpm]
Count
Rotational speed [rpm]
0
250
500
750
1000
1250
1500
1750
1250
1500
1750
2000
2250
2500
2750



Boxplots and Violinplots

The boxplot summarizes statistical values, while the violinplot also shows the estimated distribution of measurements.

```
# Vergleicht Boxplots und Violinplots der drei Messgrößen.
columns = ["Torque [Nm]", "Rotational speed [rpm]", "Air temperature [K]"]

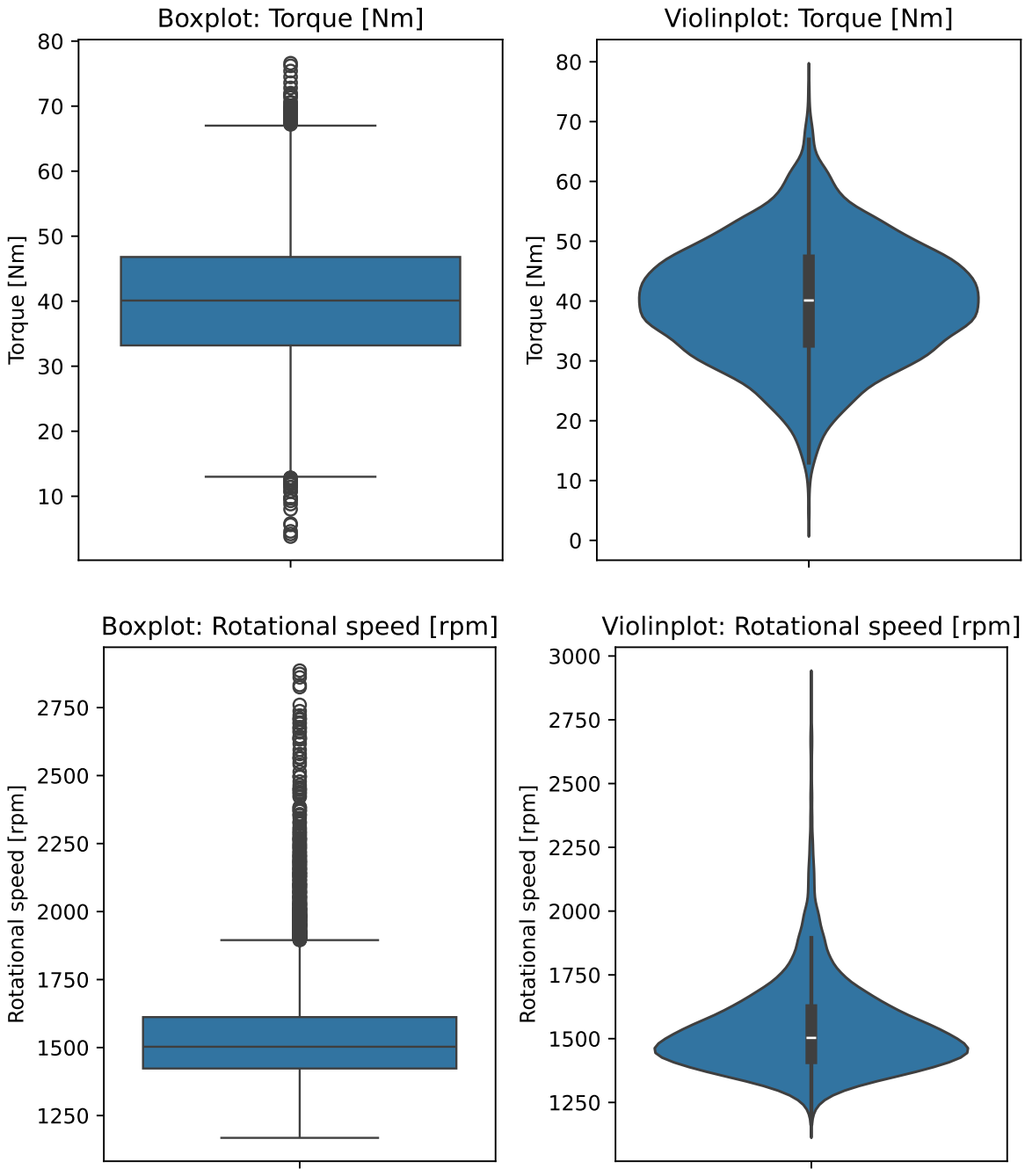
for column in columns:

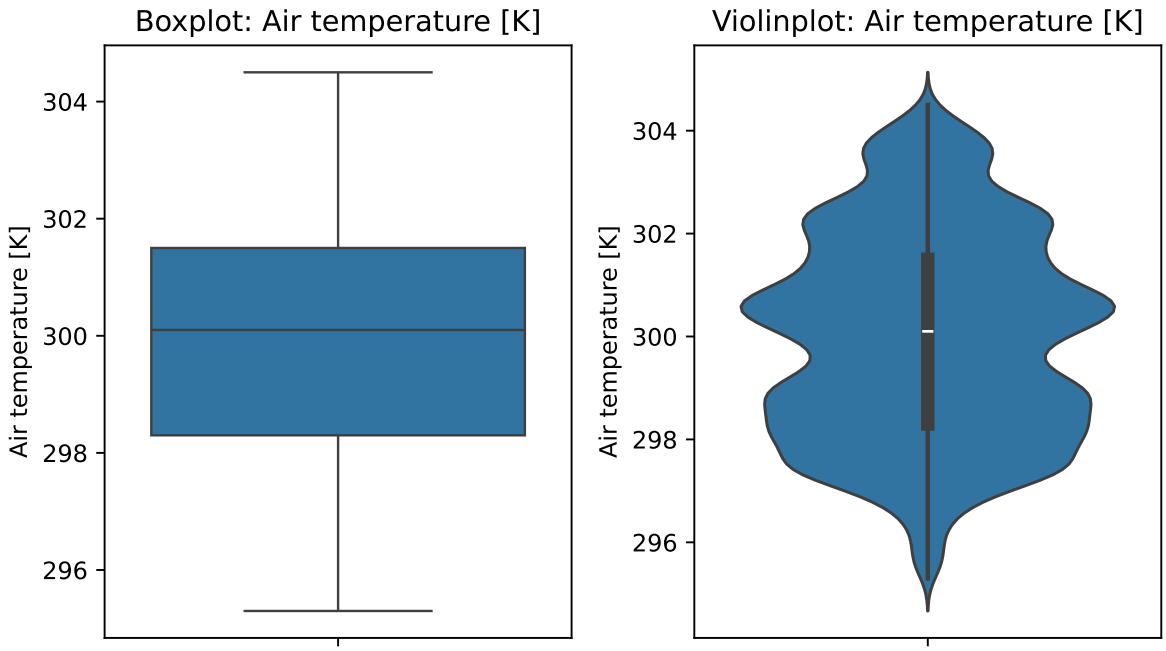
    # Erstellt zwei Diagramme nebeneinander.
    fig, axes = plt.subplots(1, 2, figsize=(7, 4))

    # Zeigt Median, Quartile, Whisker und mögliche Ausreißer.
    sns.boxplot(data=df, y=column, ax=axes[0])
    axes[0].set_title(f"Boxplot: {column}")

    # Zeigt die geschätzte Dichteverteilung.
    sns.violinplot(data=df, y=column, ax=axes[1])
    axes[1].set_title(f"Violinplot: {column}")
```

```
    plt.tight_layout()
    plt.show()
```





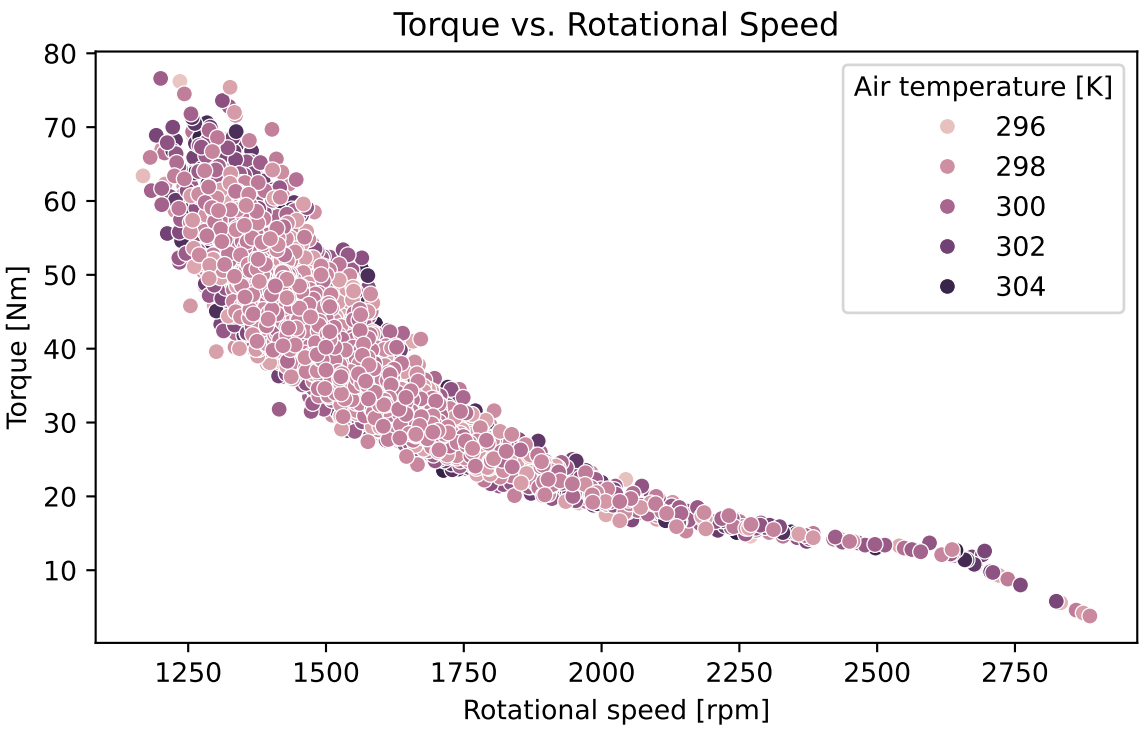
Relationships Between Variables

Scatterplot

```
# Zeigt Drehmoment und Drehzahl.
# Die Lufttemperatur wird durch die Farbe dargestellt.
plt.figure(figsize=(7, 4))

sns.scatterplot(
    data=df,
    x="Rotational speed [rpm]",
    y="Torque [Nm]",
    hue="Air temperature [K]"
)

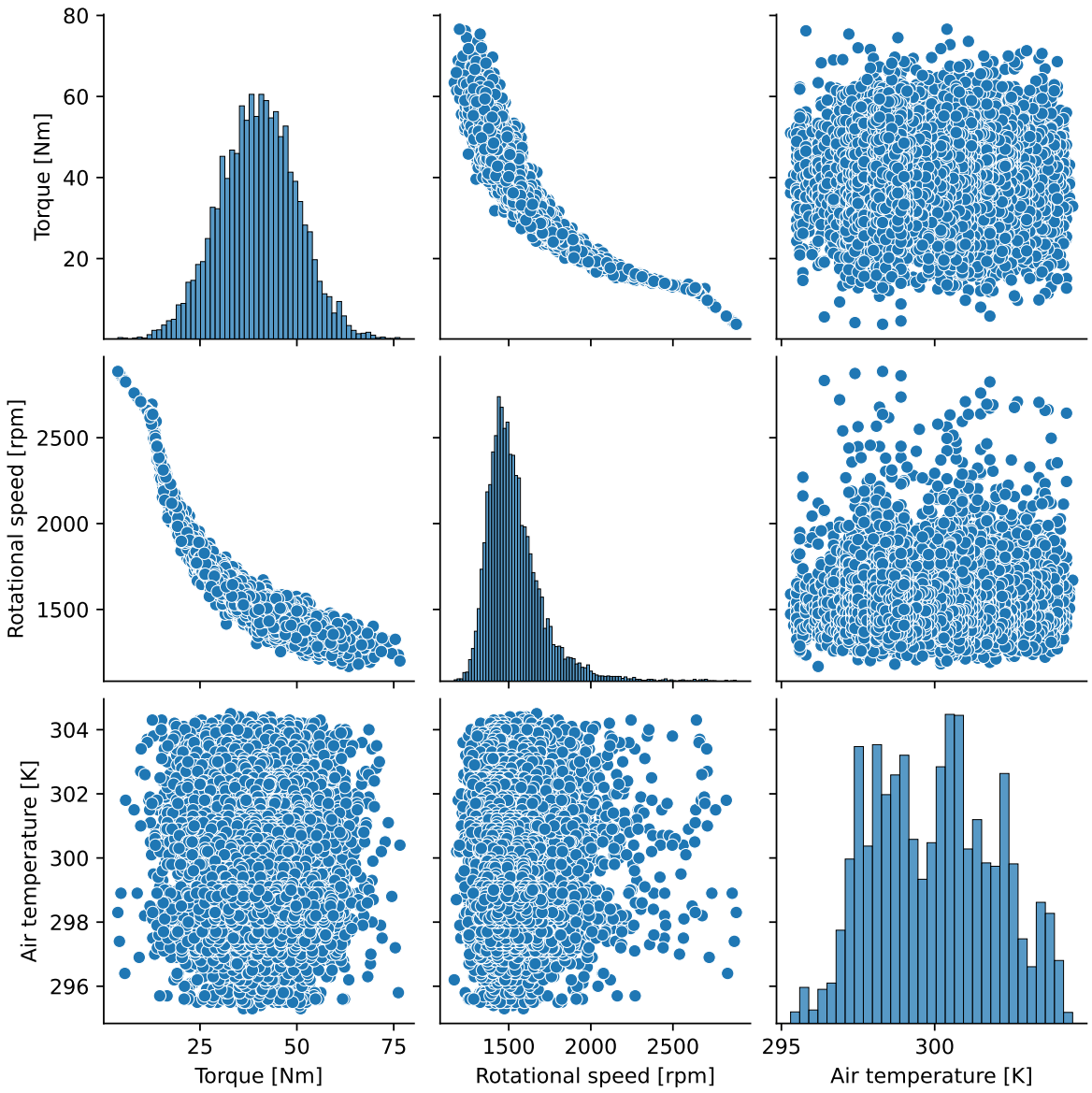
plt.title("Torque vs. Rotational Speed")
plt.xlabel("Rotational speed [rpm]")
plt.ylabel("Torque [Nm]")
plt.show()
```



Pairplot

```
# Untersucht die paarweisen Zusammenhänge aller drei Messgrößen.
columns = ["Torque [Nm]", "Rotational speed [rpm]", "Air temperature [K]"]

sns.pairplot(df[columns])
plt.show()
```



Correlation Heatmap

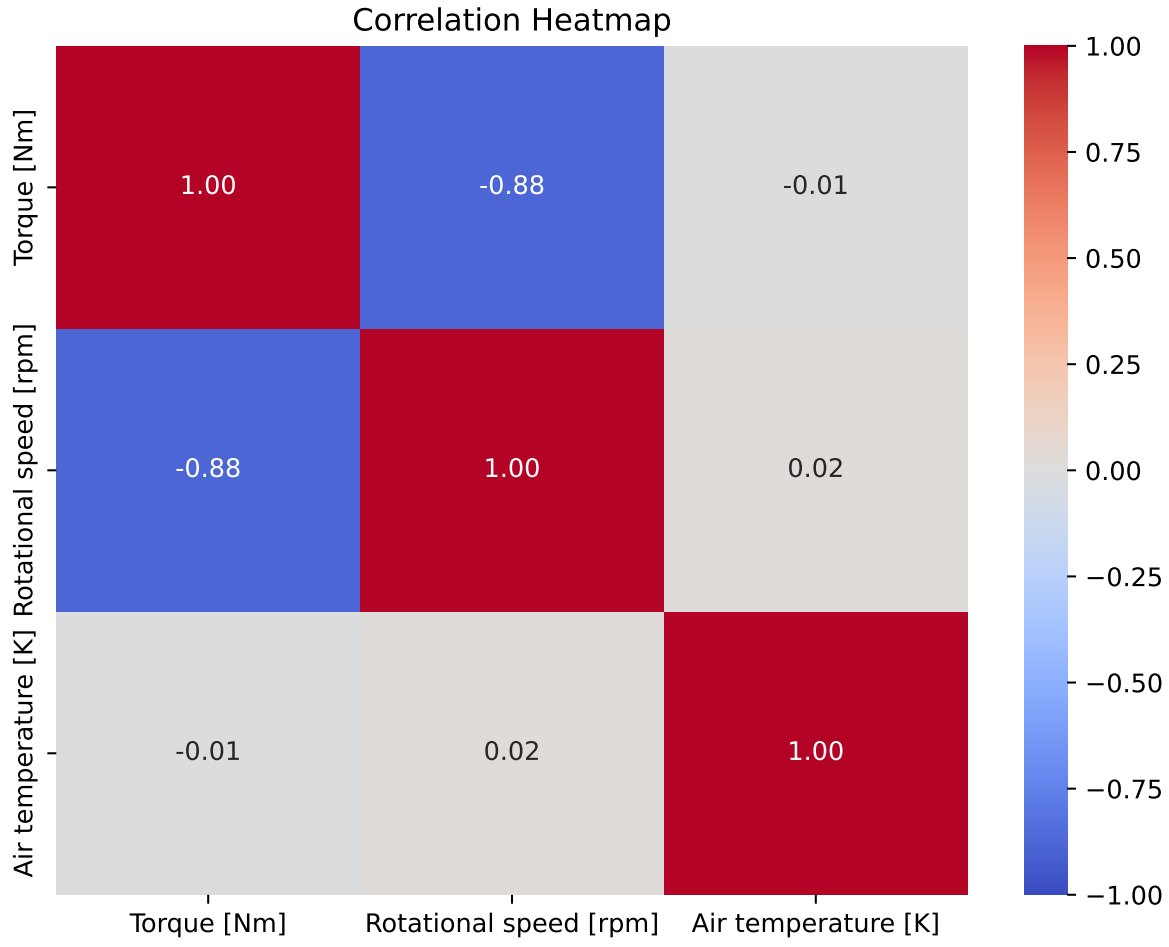
```
# Berechnet die Pearson-Korrelationsmatrix.
columns = ["Torque [Nm]", "Rotational speed [rpm]", "Air temperature [K]"]
correlation = df[columns].corr()

# Zeigt die Korrelationen als farbige Matrix.
```

```
plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    vmin=-1,
    vmax=1
)

plt.title("Correlation Heatmap")
plt.show()
```



The Pearson correlation coefficient ranges from -1 to +1. Negative values indicate a negative

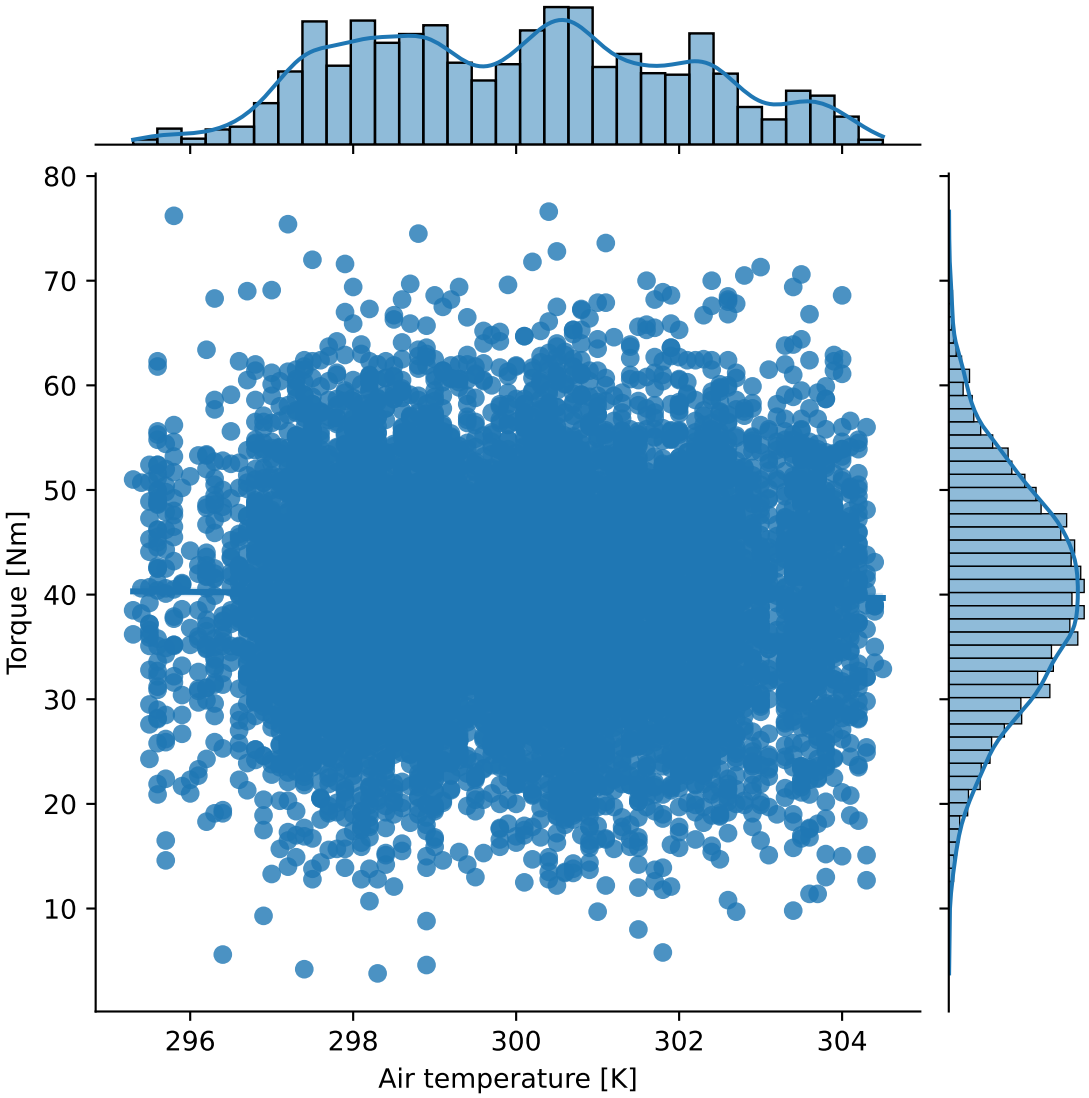
linear relationship, while positive values indicate a positive linear relationship.

Advanced Plots

Jointplot

```
# Untersucht Drehmoment und Lufttemperatur.
# kind="reg" ergänzt eine lineare Regressionsgerade.
sns.jointplot(
    data=df,
    x="Air temperature [K]",
    y="Torque [Nm]",
    kind="reg"
)

plt.show()
```



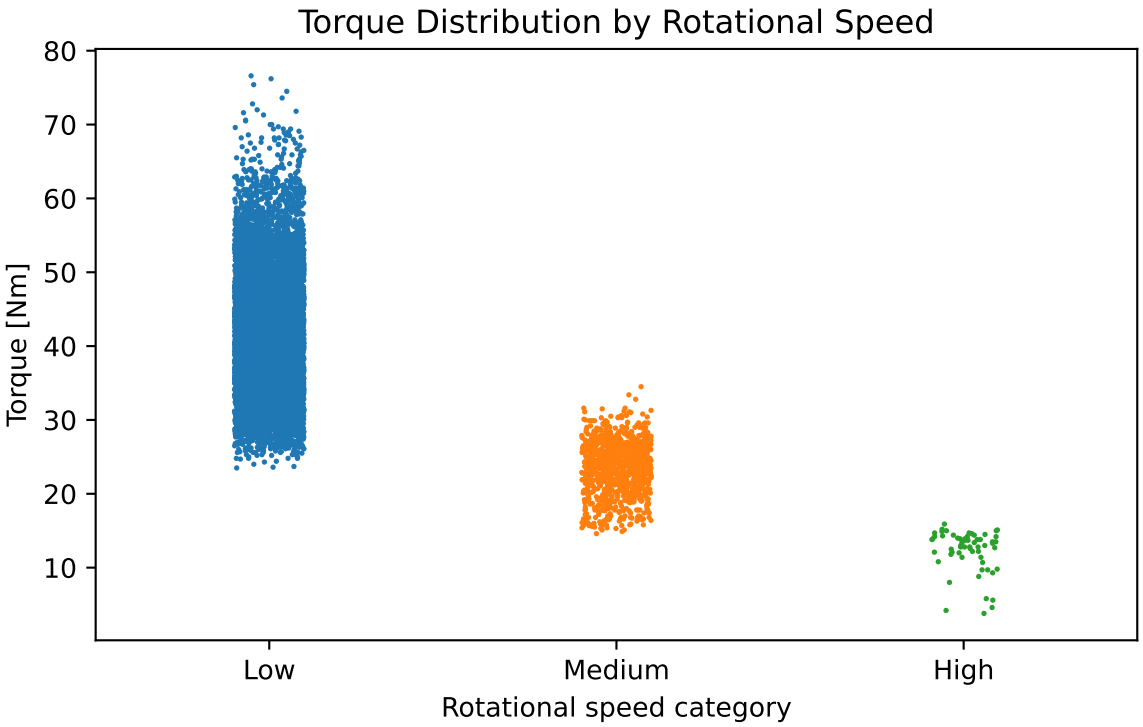
Stripplot

```
# Unterteilt die Drehzahl in drei gleich breite Bereiche.
df["Speed category"] = pd.cut(
    df["Rotational speed [rpm]"],
    bins=3,
    labels=["Low", "Medium", "High"]
)
```

```
# Zeigt die Drehmomentwerte nach Drehzahlkategorien.
plt.figure(figsize=(7, 4))

sns.stripplot(
    data=df,
    x="Speed category",
    y="Torque [Nm]",
    hue="Speed category",
    jitter=True,
    size=2
)

plt.title("Torque Distribution by Rotational Speed")
plt.xlabel("Rotational speed category")
plt.ylabel("Torque [Nm]")
plt.show()
```

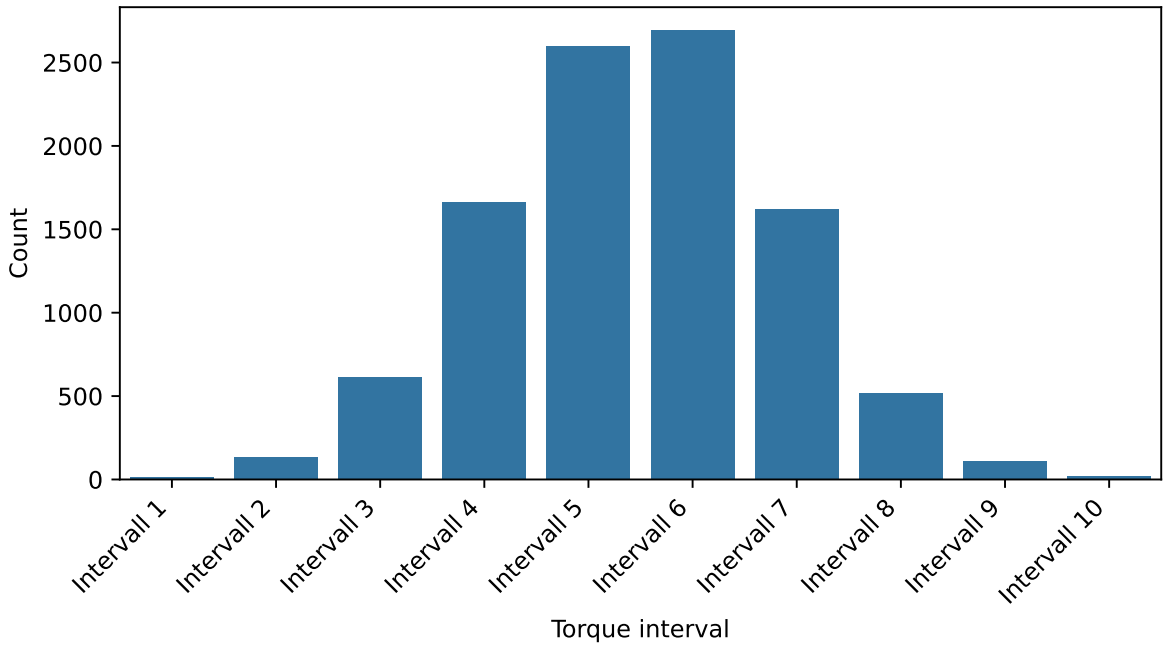


Countplot

```
# Unterteilt das Drehmoment in zehn gleich breite Intervalle.
df["Torque bin"] = pd.cut(
    df["Torque [Nm]"],
    bins=10,
    labels=[f"Intervall {i}" for i in range(1, 11)]
)

# Zählt die Messwerte in jedem Intervall.
plt.figure(figsize=(7, 4))
sns.countplot(data=df, x="Torque bin")

plt.xticks(rotation=45, ha="right")
plt.xlabel("Torque interval")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



Observations

Are there any obvious outliers in the data?

Yes, the boxplots show that especially torque and rotational speed have some outliers.

Which two variables seem most related?

Torque and rotational speed show a negative relationship.

Are there any non-linear patterns visible?

A clear nonlinear relationship is visible between torque and rotational speed. Torque decreases as rotational speed increases, but the relationship is not a straight line.

How are the distributions shaped?

Torque is approximately normally distributed, with most measurements around 40 Nm.

Rotational speed has a right-skewed distribution, with most values around 1500 rpm.

Air temperature does not show a clear normal distribution and has an irregular distribution over the measured range.